citizens and customers twenty-four hours a day. Hackers working for democracies may develop new tools to strengthen society's self-correcting mechanisms, by exposing government corruption and corporate malpractices. Both technologies could be developed.

Choice doesn't end there. Even after a particular tool is developed, it can be put to many uses. We can use a knife to murder a person, to save their life in surgery or to cut vegetables for their dinner. The knife doesn't force our hand. It's a human choice. Similarly, when cheap radio sets were developed, it meant that almost every family in Germany could afford to have one at home. But how would it be used? Cheap radios could mean that when a totalitarian leader gave a speech, he could reach the living room of every German family. Or they could mean that every German family could choose to listen to a different radio programme, reflecting and cultivating a diversity of political and artistic views. East Germany went one way; West Germany went the other. Though radio sets in East Germany could technically receive a wide range of transmissions, the East German government did its best to jam Western broadcasts and punished people who secretly tuned in to them.[55] The technology was the same, but politics made very different uses of it.

The same is true of the new technologies of the twenty-first century. To exercise our agency, we first need to understand what the new technologies are and what they can do. That's an urgent responsibility of every citizen. Naturally, not every citizen needs a PhD in computer science, but to retain control of our future, we do need to understand the political potential of computers. The next few chapters, then, offer an overview of computer politics for twenty-first-century citizens. We will first learn what the political threats and promises are of the new computer network and will then explore the different ways that democracies, dictatorships and the international system as a whole might adjust to the new computer politics.

Politics involves a delicate balance between truth and order. As computers become important members of our information network, they are increasingly tasked with discovering truth and maintaining order. For example, the attempt to find the truth about climate change increasingly depends on calculations that only computers can make, and the attempt to reach social consensus about climate change increasingly depends on recommendation algorithms that curate our news feeds, and on creative algorithms that write news stories, fake news and fiction. At present, we are